

The integral test.

Ex. Test the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ for convergence or divergence.

Sln. let $f(x) = \frac{1}{x^2+1}$ is continuous and positive on $[1, \infty)$ and

$f(x)$ is decreasing on $[1, \infty)$. Indeed,

$$f'(x) = -\frac{1}{(x^2+1)^2} (2x) = \frac{-2x}{(x^2+1)^2} < 0 \text{ on } [1, \infty).$$

By the integral test:

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^2+1} dx &= \lim_{b \rightarrow \infty} \arctan x \Big|_1^b = \lim_{b \rightarrow \infty} \arctan b - \arctan(1) \\ &= \frac{\pi}{2} - \frac{\pi}{4} \text{ ft.} \end{aligned}$$

So, $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ is convergent.

$$\text{is } \sum_{n=1}^{\infty} \frac{1}{n^2+1} = \frac{\pi}{2} - \frac{\pi}{4}.$$

Example: (p -test). For what values of p , $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent?

Sln: If $p < 0$ $\frac{1}{n^p} = n^{-p}$ and $\lim_{n \rightarrow \infty} \frac{1}{n^p} = \infty$ so by divergent

test $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent.

if $p > 0$ then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$. let $f(x) = \frac{1}{x^p}$ is continuous, positive, and decreasing so.

$\int_1^{\infty} \frac{1}{x^p} dx$ is converges if $p > 1$ and diverges if $p \leq 1$.

Hence,

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ is } \begin{cases} \text{convergent if } p > 1 \\ \text{divergent if } p \leq 1. \end{cases}$$

p -test for series.

Example: The series $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent b/c is a p-series where $p=3 > 1$.

and $\sum_{n=1}^{\infty} \frac{1}{n^{1/3}}$ is divergent b/c is a p-series where $p=\frac{1}{3} < 1$.

Rmk: We should not infer from the integral test that the sum of the series is equal to the value of the integral, i.e.,
$$\sum_{n=1}^{\infty} a_n \neq \int_1^{\infty} f(x) dx.$$

Estimating the sum of a series:

Suppose that $a_k = f(k)$, where f is continuous, positive, decreasing, and $\sum_{n=1}^{\infty} a_n$ is convergent then

$$\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx. \quad (71)$$

Where $R_n = a_{n+1} + a_{n+2} + a_{n+3} + \dots$

Example: Use (71) with $n=10$ to estimate the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$.

Sln:
$$\int_{10+1}^{\infty} \frac{1}{x^3} dx \leq \sum_{k=10+1}^{\infty} \frac{1}{k^3} \leq \int_{10}^{\infty} \frac{1}{x^3} dx. \quad \text{So.}$$

$$S_{10} + \int_{10+1}^{\infty} \frac{1}{x^3} dx \leq \sum_{k=1}^{\infty} \frac{1}{k^3} \leq S_{10} + \int_{10}^{\infty} \frac{1}{x^3} dx.$$

$$\int_{11}^{\infty} \frac{1}{x^3} dx = \frac{1}{2(11)^2} \quad \text{and} \quad \int_{10}^{\infty} \frac{1}{x^3} dx = \frac{1}{2(10)^2}. \quad S_{10} \approx 1.197532.$$

$$\text{then, } 1.197532 + \frac{1}{2(11)^2} \leq S \leq 1.197532 + \frac{1}{2(10)^2}.$$

$$1.201664 \leq s \leq 1.202532.$$